

20 Which statement about benzene and its properties is correct?

- A** Benzene undergoes nucleophilic substitution reactions with Br_2 in the presence of AlBr_3 .
- B** The benzene molecule is planar so hydrogen can easily undergo addition reactions with it without the use of a catalyst.
- C** The π electrons in benzene are able to donate an electron pair to attack a carbocation to form a bond.
- D** The sideways overlap of p orbitals in benzene means the C–C bonds alternate between long, single bonds and short, double bonds.